



EAST WEST UNIVERSITY
Department of Mathematical & Physical Sciences

Quiz-I, Sec 6, Summer 2025 Semester

Course:	PHY 109 Engineering Physics – I
Instructor:	Md. Mahfuzul Haque, Assistant Professor, Department of MPS
Full Marks:	10
Time:	30 Minutes
Name:	
ID:	

Notes:

Answer ALL of them. Mark of each question is mentioned at the right margin.

1. In simple harmonic motion, how is the acceleration of an object related to its displacement from the equilibrium position? What this relation reveals about the motion. [Mark: 2]
2. Imagine you are observing a smooth horizontal surface where a block of mass m is attached to a spring and is oscillating back and forth in a simple harmonic motion. The maximum displacement from the equilibrium position is A .
 - (a) Write the expression for the displacement $x(t)$ of the block as a function of time and mention each term of the expression. [Mark: 1]
 - (b) From the displacement, derive the expression for the velocity $v(t)$. [Mark: 1]
 - (c) Use the velocity to find the kinetic energy of the block. [Mark: 1]
 - (d) Determine the maximum kinetic energy. [Mark: 1]
3. You're working with a spring-block system on a frictionless horizontal surface. A block of mass 520 g is attached to a spring with a spring constant of 80 N/m. You pull the block 9 cm away from its equilibrium position and release it from rest at $t = 0$.
 - (a) What is the amplitude of oscillation? [Mark: 1]
 - (b) Determine the time period of oscillation. [Mark: 2]
 - (c) As the block swings back and forth, it reaches its top speed at a certain point. Determine the maximum speed of the block during the motion? [Mark: 1]